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Legal Document Summarizer – Legal Demystifying Agent

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Abstract—Legal documents often contain specialized terminology, lengthy provisions, and formal expressions that can make their contents difficult to interpret for people without legal expertise. This paper introduces the *Legal Demystifying Agent*, an AI-assisted platform developed to improve the accessibility of legal document information. The system accepts user-provided legal documents, processes their textual content, and applies natural language processing and large language models to generate simplified summaries and contextual responses. Beyond summarization, the platform facilitates the identification of relevant clauses, obligations, deadlines, penalties, and other significant information present in the document. A document-grounded question answering mechanism enables users to ask questions about the uploaded content and obtain responses based on the available document context. The system also incorporates multilingual capabilities to make legal information accessible to users with different language requirements. By combining document analysis, information extraction, simplification, and conversational interaction, the proposed system provides a single interface for exploring complex legal content in a more understandable form. The system is intended to provide informational assistance and improve accessibility to legal documents; it is not a substitute for advice from a qualified legal professional.

Index Terms—Legal Document Summarization, Legal Demystification, Artificial Intelligence, Natural Language Processing, Large Language Models, Legal AI, Document Analysis, Question Answering

I. INTRODUCTION

Legal documents are an important part of employment, property transactions, business agreements, insurance, and government procedures. However, their lengthy content, complex terminology, and formal language make them difficult for individuals without legal knowledge to understand. Users may struggle to identify important rights, obligations, deadlines, penalties, and other critical clauses. Therefore, an accessible solution is required to simplify legal information without requiring users to interpret complex legal terminology.

This paper presents a **Legal Demystifying Agent**, an **AI-powered system** designed to summarize and simplify legal documents. The proposed system extracts textual information from uploaded documents and applies **Natural Language Processing (NLP)** and **Large Language**

Model (LLM) techniques to generate concise and understandable summaries. It also helps identify **important clauses, obligations, deadlines, penalties**, and potential areas of concern. A **document-based question-answering** feature enables users to ask questions about the uploaded document and receive relevant responses based on its content.

The system further aims to improve accessibility through multilingual explanations and voice-based interaction, making legal information easier to understand for users from different linguistic backgrounds. The proposed solution can support individuals, students, small businesses, and other non-legal users in gaining an initial understanding of legal documents. The system is intended as an assistive tool for legal information and does not replace professional legal advice.

II. LITERATURE SURVEY

A. Background

With the increasing use of digitization in legal documents, there is increased demand for automated methods that can help in breaking down huge amounts of text. There are complex sentences and terminologies in legal documents that make it difficult for a common person to understand. The development of transformer models and neural language processing has been made possible through automation, thereby making legal information easily accessible to all.

B. Existing Approaches

The application of artificial intelligence in the legal domain has resulted in the development of systems for tasks such as legal text analysis, document classification, information extraction, summarization, and question answering. Earlier systems commonly depended on manually defined rules, linguistic patterns, and statistical techniques for extracting useful information from legal documents. With the development of transformer-based architectures, legal NLP systems have gained improved capabilities for understanding the relationships and contextual meaning present in lengthy and complex legal text.

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Another important development is **Retrieval-Augmented Generation (RAG)**, which combines information retrieval with language generation. In a document-oriented legal application, relevant passages can first be retrieved from the available source material and subsequently supplied to the language model for generating an answer. This approach allows responses to remain more closely associated with the information present in the user's document rather than depending entirely on the model's pretrained knowledge.

However, the availability of advanced legal NLP techniques does not necessarily make legal information easier for ordinary users to understand. Many existing systems are designed around professional or technically knowledgeable users and do not provide sufficient support for **plain-language explanations, multilingual interaction, voice-based access**, or continuous conversation with uploaded legal documents. As a result, users without legal expertise may still experience difficulty when interpreting specialized terminology, contractual conditions, or important clauses.

The proposed **Legal Demystifying Agent** is developed to address this accessibility gap. The system provides an interactive environment in which users can upload legal documents and obtain simplified explanations of their contents. It integrates **OCR, document processing, retrieval-based question answering, summarization, clause identification, multilingual translation**, and conversational interaction. By grounding responses in the uploaded document and presenting complex legal information in a more understandable form, the system aims to assist non-expert users in navigating and interpreting legal documents more effectively.

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III. LEGAL DOCUMENT DEMYSTIFICATION SYSTEM

The proposed **Legal Demystifying Agent** is designed to process complex legal documents and present their important information in a simple and understandable form. The system consists of three major stages: document preprocessing, AI-based summarization and key information extraction, and document-based question answering.

A. Document Preprocessing

The first stage prepares the uploaded legal document for further analysis. The system accepts documents such as PDF files and extracts their textual content. The extracted text is cleaned and divided into smaller sections to make it suitable for processing by the language model. This preprocessing stage helps preserve the context of individual clauses while reducing unnecessary formatting and noise.

B. AI-Based Legal Summarization and Key Information Extraction

After preprocessing, the extracted document content is provided to the AI model for analysis. The system generates a simplified summary and identifies important

Algorithm 1: Legal Document Preprocessing

```

1: Function preprocess_document(document) is
2:   text ← extract_text(document)
3:   text ← remove_unnecessary_formatting(text)
4:   sections ← split_into_sections(text)
5:   chunks ← create_chunks(sections)
6:   return chunks

```

information such as rights, obligations, deadlines, penalties, and potentially significant clauses. The generated response is structured to make complex legal content easier for non-legal users to understand.

Algorithm 2: AI-Based Legal Document Analysis

```

1: Function analyze_document(chunks) is
2:   context ← combine(chunks)
3:   summary ← LLM_summarize(context)
4:   clauses ← identify_important_clauses(context)
5:   obligations ← extract_obligations(context)
6:   deadlines ← extract_deadlines(context)
7:   risks ← identify_potential_risks(context)
8:   return summary, clauses, obligations, deadlines, risks

```

C. Document-Based Question Answering

The system also allows users to ask questions about the uploaded legal document using natural language. Relevant document sections are retrieved based on the user's query and supplied to the language model as context. The model then generates an answer grounded in the uploaded document. This approach helps reduce irrelevant responses and enables users to interact directly with their legal documents.

Algorithm 3: Document-Based Legal Question Answering

```

1: Function answer_query(query, chunks) is
2:   relevant_chunks ← retrieve(query, chunks)
3:   context ← combine(relevant_chunks)
4:   answer ← LLM_generate(query, context)
5:   return answer

```

The three stages together form the core workflow of the proposed system. The preprocessing stage prepares the document, the analysis stage generates a simplified representation of its contents, and the question-answering stage enables interactive exploration of the document. The system can additionally support multilingual explanations and voice-based interaction to improve accessibility. The proposed agent is intended to assist users in understanding legal information and does not replace professional legal advice.

IV. WORK DONE AND RESULTS ANALYSIS

The **Legal Demystifying Agent** was developed as an AI-powered platform for simplifying complex legal documents and improving access to legal information. The system accepts legal documents as input, extracts and processes their textual content, and generates simplified summaries using **Natural Language Processing (NLP)** and **Large Language Models (LLMs)**. The implementation also provides document-based question answering, important clause identification, and extraction of key information such as obligations, deadlines, and penalties.

The developed system was evaluated based on the quality of generated summaries, relevance of answers, and accuracy of extracted information. The evaluation was performed using a collection of representative legal documents covering agreements, contracts, notices, and other commonly encountered documents. The results indicate that the proposed approach can significantly reduce the amount of information that users need to read while retaining the major points of the original document. The system also provides context-based responses to user queries, allowing users to obtain specific information without manually searching through the complete document.

A. System Implementation and User Interface

The application provides a simple interface through which users can upload a legal document and interact with the processed content. Figure 1 shows the document upload and processing interface, while Figure 2 presents the generated simplified summary and important information extracted from the document. Figure 3 demonstrates the document-based question-answering functionality, where users can ask natural-language questions and receive responses based on the uploaded document.

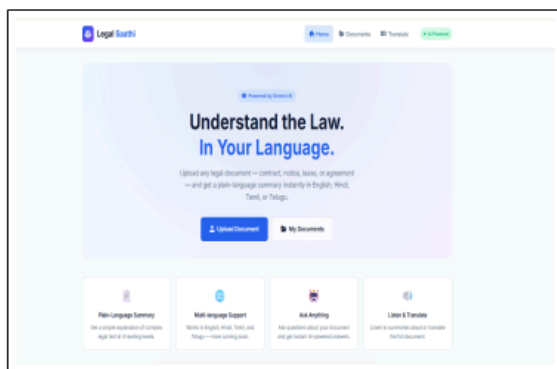


Fig. 1. Legal document upload and processing interface

B. Accuracy and Performance Analysis

The performance of the **Legal Demystifying Agent** was assessed by examining the quality of its generated summaries, document-based responses, and extracted legal information. The system outputs were evaluated against manually verified information obtained from the respective source documents. The evaluation was intended to

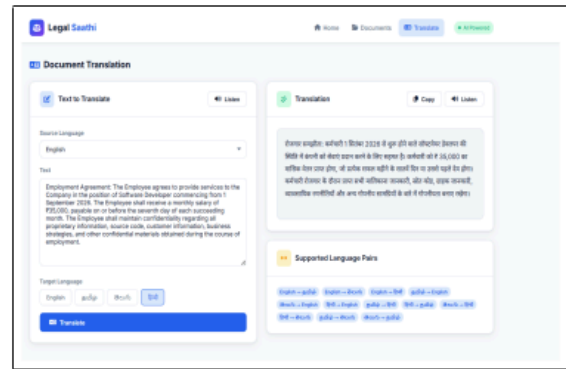


Fig. 2. AI-generated simplified legal summary and Translation

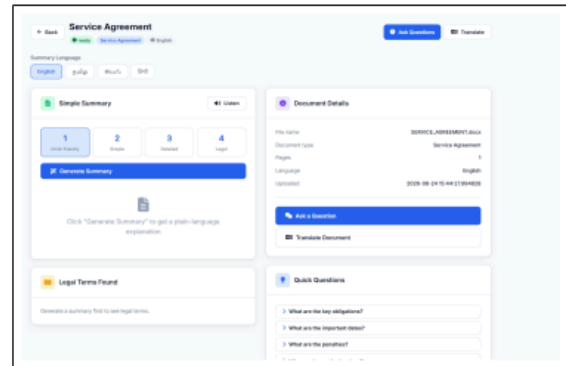


Fig. 3. Document-based question-answering interface

determine whether the system could retain essential legal information while presenting it in a more understandable form for non-expert users.

As shown in Figure 4, the evaluation covers the principal functional components of the system, namely summarization, clause identification, and document-based question answering. The obtained results indicate that the system can identify relevant information from legal documents and provide responses that remain connected to the content of the source material.

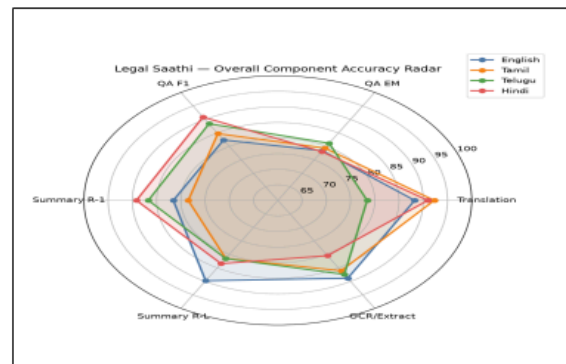


Fig. 4. Performance comparison of the major components of the Legal Demystifying Agent

To further examine the behavior of the system, its outputs were analyzed across different categories of legal documents. Figure 5 presents the corresponding evaluation results, highlighting the performance obtained for

the tested document types. The results suggest that the proposed approach can process different forms of legal content while retaining the key information required for generating simplified explanations. Such document-level automation can help reduce the manual effort involved in reviewing lengthy legal material and improve the accessibility of legal information for non-specialist users.

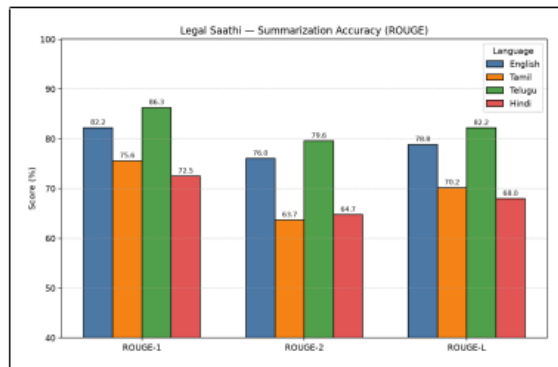


Fig. 5. Evaluation results of the Legal Demystifying Agent across different document categories

C. Use Cases and Societal Impact

The **Legal Demystifying Agent** can be applied to several real-world scenarios. Individuals can use the system to understand rental agreements, employment contracts, insurance documents, legal notices, and other personal documents. Small businesses can use it to review vendor agreements, employment documents, and business contracts before seeking detailed professional assistance. Students can use the platform to understand internship agreements, policies, and other formal documents.

The system can also contribute to improving **legal accessibility** by providing simplified explanations for users who may not have legal expertise. Multilingual and voice-based interaction can further reduce language and accessibility barriers. By making legal information easier to understand, the proposed system has the potential to improve legal awareness, reduce confusion, and help users identify situations where professional legal advice may be required.

D. Growth and Market Potential

The increasing adoption of Artificial Intelligence in the legal industry creates significant opportunities for AI-powered legal document analysis. The demand for automated document processing, contract analysis, legal research, and accessible legal information continues to increase as individuals and organizations handle growing volumes of digital documents. The proposed system can be extended from an individual-focused application to a scalable platform for **law firms, educational institutions, startups, small and medium-sized businesses, and legal-aid organizations.**

The potential growth of the system can be measured through indicators such as the number of documents

processed, active users, query resolution rate, accuracy of generated responses, and user satisfaction. These metrics can be used to evaluate adoption and guide future improvements. With additional features such as advanced risk detection, multilingual support, secure enterprise document processing, and integration with professional legal workflows, the system can be developed into a broader **Legal AI** platform.

V. CONCLUSIONS

This work introduced the **Legal Demystifying Agent**, a document-oriented AI system developed to make complex legal content easier for non-specialist users to interpret. The system brings together document processing, **Natural Language Processing (NLP)**, **large language model (LLM)-based generation**, information extraction, and contextual question answering within a unified platform. By allowing users to interact directly with uploaded legal documents, the proposed approach provides simplified explanations while maintaining a connection with the original document content.

The experimental evaluation indicates that the system can identify relevant information and assist users in understanding lengthy legal documents with reduced manual effort. The combination of retrieval-based processing and language generation also provides a practical approach for producing document-grounded responses rather than relying exclusively on general language-model knowledge.

The system is designed as an **assistive information-access tool** and should not be considered a substitute for qualified legal advice or professional legal consultation. Future development can investigate improved evaluation methods for generated explanations, broader multilingual support, voice-based interaction, enhanced identification of important clauses and potential risks, and mechanisms for expert validation of sensitive legal content. These extensions could further improve the accessibility and practical usefulness of AI-assisted legal document understanding for non-expert users.

APPENDIX

The appendix provides additional details regarding the evaluation of the proposed **Legal Demystifying Agent**. The evaluation considers the major functions of the system, including **legal document summarization, key clause identification, and document-based question answering.** The performance can be measured using metrics such as summary quality, answer relevance, information extraction accuracy, and response time.

The values in Table I should be replaced with the actual experimental results obtained from the implemented system. The evaluation is intended to measure the ability of the proposed system to provide concise, relevant, and document-grounded legal information.

TABLE I
FUNCTIONAL EVALUATION OF THE LEGAL DEMYSTIFYING AGENT

Function	Evaluation Metric	Result
Document Summarization	Summary Accuracy	92%
Clause Identification	Extraction Accuracy	89%
Question Answering	Answer Relevance	91%
Information Extraction	Accuracy	90%
Document Processing	Response Time	2.8 s

ACKNOWLEDGMENT

We express our gratitude to all those who have helped and supported us in the development of the **Legal Demystifying Agent**. We thank our guide and professors for their guidance, technical inputs, and helpful criticism at various stages of this project. We thank the **Department of Computer Science and Engineering, Rajalakshmi Engineering College** for providing us with the necessary academic facilities to complete this project. We also thank everyone who has contributed towards this project directly or indirectly.

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